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Total No. of printed pages = 04

Monsoon, 2025

B. Tech 3rd Semester Examination

SOLID MECHANICS

Course Code: BCE23301T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

Q.1 Answers all the questions

1×10=10

(i) Which of the following is a dimensionless quantity

- (a) Shear Stress
- (b) Poisson's Ratio
- (c) Strain
- (d) Both (b) and (c)

(ii) The relationship between Young's modulus (E), Bulk modulus (K) and Poisson's ratio (μ) is given by [CO1]

- (a) $E=2K(1-2\mu)$
- (b) $E=3K(1-2\mu)$
- (c) $E=2K(1-3\mu)$
- (d) $E=5K(1-3\mu)$

(iii) The Planes, which have no shear stress are known as-

- (a) Tangential Planes
- (b) Nominal Planes
- (c) Normal Planes
- (d) Principal Planes

(iv) The angle between Major principal Plane and Minor Principal Plane for a strained body is-

- (a) 45°
- (b) 30°
- (c) 60°
- (d) 90°

(v) The maximum value of shear stress in the thin cylindrical vessel of internal diameter d , thickness t and subjected to internal pressure p is given by-

(a) $\tau = \frac{pd}{4t}$

(b) $\tau = \frac{pd}{2t}$

(c) $\tau = \frac{pd}{8t}$

(d) $\tau = \frac{pd}{t}$

(vi) In thin cylindrical vessel, the circumferential stress has the highest value at-

(a) Centre

(b) Inner surface

(c) outer surface

(d) middle of the thickness

(vii) Which of the following is the **CORRECT** statement for a cantilever beam-

(a) Deflection is maximum at the fixed end

(b) Deflection is maximum at the free end

(c) Slope is maximum at the fixed end

(d) Both slope and deflection is maximum at the free end

(viii) A long thin-walled cylindrical shell closed at both ends, is subjected to an internal pressure. Find the ratio of hoop stress to longitudinal stress is-

(a) 0.25

(b) 0.5

(c) 1

(d) 2

(ix) A motor driving a solid circular shaft transmit 40 kW of power at 500 rpm. If the diameter of the shaft is 40 mm, the maximum shear stress in the shaft is-

(a) 60.85 MPa

(b) 3.8 MPa

(c) 30.42 MPa

(d) 60.85 Pa

(x) In cantilever beams, the deflection is zero at _____

(a) Free end

(c) At supports

(b) Fixed end

(d) Through out

Q.2.

(i) Draw the stress strain diagram for Mild Steel under tension and show the important points and region.

(ii) An axial pull of 35000 N is acting on a bar consisting of three lengths as shown in the Figure 1 below. If the Young's modulus = $2.1 \times 10^5 \text{ N/mm}^2$, determine:

(a) Stresses in each section

(b) Total extension of the bar

3+7=10

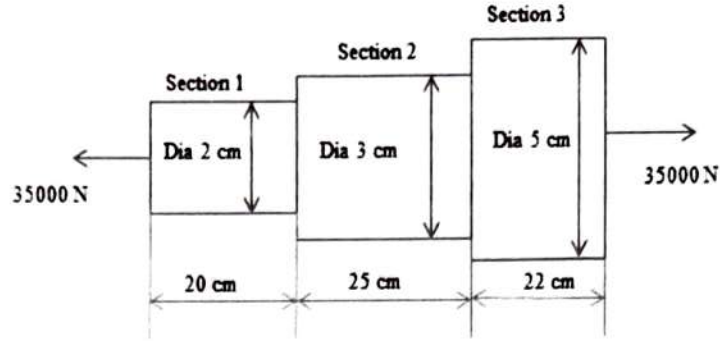


Figure-1

OR

(iii) What is strain energy? Derive the expression for strain energy. 5

(iv) Derive the expression of volumetric strain. 5

Q.3

(i) What do you mean by principal plane and principal stress? Briefly discuss. 4

(ii) Find the normal and shear stresses on the oblique plane shown in the Figure 2 below. 6

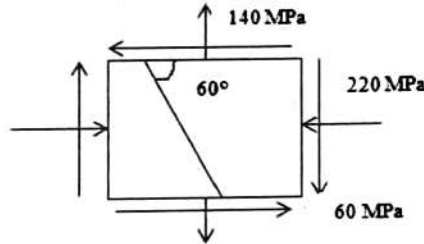


Figure-2

OR

(iii) Find Graphically from Mohr's cycle the principal stresses and the maximum shear stress for the plane stress case $\sigma_x = 300$ MPa, $\sigma_y = 800$ MPa, $\tau_{xy} = 300$ MPa. Check your answers analytically. 10

Q.4

(i) Derive the torsion equation $\frac{T}{J} = \frac{\tau}{R} = \frac{G\theta}{L}$ 10

OR

(ii) State the assumptions in simple bending theory. 5

(iii) A thin cylindrical shell 3 m long, 1.2 m in diameter is subjected to an internal pressure of 1.67 N/mm². If the thickness of the shell is 13 mm, find the circumferential and longitudinal stresses. Also find the maximum shear stress. Take $E = 2 \times 10^5$ N/mm² and Poisson's ratio = 0.28. 5